Mustapha Muhsin

AI BACKEND ENGINEER

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Results-driven AI Backend Engineer specializing in cloud-native serverless systems, database integrations, and custom agentic loop orchestrations. Proven experience designing scalable backend APIs using TypeScript and Python, deploying AWS workloads (from Lambdas to containerized ECS/Fargate services), and building multi-agent workflows with tool-calling capabilities. Focuses on delivering immediate business efficiency by replacing manual operations with optimized data and automation pipelines.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++

Backend & Databases: FastAPI, Node.js, Express, PostgreSQL, MySQL, DynamoDB, Redis, Drizzle ORM, RESTful APIs, Server-Sent Events (SSE)

AWS & Cloud Infrastructure: AWS (Lambda, ECS, Fargate, Step Functions, DynamoDB, S3, IAM, API Gateway, Cognito, Bedrock), Railway, Docker, Vercel

AI Engineering: Governed Agent Loops, Harness Engineering, Multi-Agent Swarm Orchestration, RAG Pipelines, Tool-Calling / Function Binding, LangChain, LangGraph

Developer Tools: Git, Docker, Cursor, Claude Code, GitHub Actions, AWS CLI

EXPERIENCE

White Label Resell — Remote - Los Angeles, CA

AI Backend Engineer — *January 2025 - Present*

Governed Agent Swarm & Loop Harness Architecture

- Architected a 5-phase governed agentic build loop combining read-only Pi swarm agents for codebase analysis, confidence scoring reconciliation, write-enabled implementer sessions, and Steel.dev Playwright E2E proofing before GitHub PR creation.

SEO Content AI (SCAI) — AI Content Generation Platform

- Implemented the parallel article generation architecture — building the orchestrator, modular generators for all 9 formats, and a deterministic assembler — enabling concurrent article production at scale for the platform release and AI Site Builder integration.
- Cut bulk generation time by 90% via a fair concurrency system scaling to 40 simultaneous articles with real-time progress streaming.
- Built the Stripe billing infrastructure from scratch — atomic credit transactions, monthly plan allocations, per-component billing accuracy, and a billing UI — making the platform commercially operational.
- Eliminated generation loss from browser interruptions by building a hybrid session persistence system that automatically reconnects and resumes in-progress jobs.

Enterprise Waste-Operations Platform (under NDA)

- Designed the AWS serverless architecture for a role-based waste operations platform (POC) to replace a manual Excel workflow, enabling automated uploads, role-based access, and Power BI reporting for ~100 users.
- Produced all technical deliverables: an interactive system architecture diagram, a Technical Brief (VPC isolation, DB partitioning, API specs), a cloud cost calculator (AWS/Azure), and a vendor comparison matrix — reviewed by the client's IT Director.

Client Lead Automation

- Replaced a paid lead capture tool with a custom automation that intercepts client form submissions, enriches leads with traffic attribution (UTM, Google Ads, referrer), syncs to Google Sheets and GA4, and sends weekly reports — eliminating recurring costs.

National Centre for AI & Robotics (NCAIR) — NITDA — Abuja, Nigeria

AI & Embedded Systems Intern — *February 2022 - July 2022*

- Built the hardware prototype for Elonosia AI (an NCAIR-incubated startup for ML-based malaria detection) — designing the automated microscopy capture device with ESP32/ESP8266, Altium PCBs, and 3D-printed enclosures to bridge the model to clinical use.
- Contributed to training and validation of TensorFlow CNN models for blood smear malaria detection.
- Delivered hands-on STEM workshops for children through NITDA's SB4Kids program — using 3D-printed Otto robots and microcontrollers to introduce IoT, robotics, and programming.

EDUCATION

Federal University of Technology Minna

B.Eng., Electrical and Electronics Engineering